

ARNOLD ABRAHAM

SOFTWARE ENGINEER | BACKEND | DATA & AI

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Location: Erlangen, Germany (Willing to relocate)

LinkedIn: [linkedin.com/in/arnoldabraham](https://www.linkedin.com/in/arnoldabraham) GitHub: github.com/arnold-abraham



Experience

Software Engineer – Working Student

Jul 2023 – Jun 2025

Siemens Logistics (Vanderlande), Nuremberg, Germany

- Built a Python-based predictive maintenance pipeline integrating thermal sensor data and ML models, enabling early fault detection and reducing periodic manual inspections by 90%.
- Led the migration of two Java applications from Oracle to PostgreSQL, ensuring data integrity and application stability while reducing database licensing costs.
- Developed backend features and implemented HikariCP connection pooling to improve performance and stability.
- Designed and implemented automated installation workflows for JRE, PostgreSQL, and environment configuration, reducing setup effort by 70%.
- Implemented an asyncio-based Python data acquisition system for concurrent real-time monitoring of industrial drives.
- Built and maintained CI/CD pipelines in GitLab CI; managed containerised deployments using Docker and AWS.

Product Engineer

Sept 2021 – Sept 2022

IBS Software, Kochi, India

- Developed Java/Spring-based backend components and RESTful APIs for an enterprise airline crew scheduling platform used by LATAM Airlines and Qantas, supporting thousands of daily scheduling decisions.
- Designed and implemented complex scheduling and compliance rules, addressing production issues and delivering enhancements based on QA and operational feedback.
- Created and troubleshooted SQL scripts across relational systems (PostgreSQL, Oracle) to support backend functionality and ensure data integrity.
- Worked across the full SDLC in an agile environment, including unit and integration testing using JUnit and Mockito, peer code reviews, and production support.

Education

MSc Data Science

Oct 2022 – Sept 2025

University of Erlangen–Nuremberg, Germany

- Major in Artificial Intelligence and Machine Learning.
- Thesis: *Predictive Maintenance for Electrical Panels: Hotspot Forecasting and Anomaly Detection* (Siemens).

B.Tech Computer Science and Engineering

Aug 2017 – Jul 2021

APJ Abdul Kalam Technological University, India

Technical Skills

Languages: Java, Python (asyncio, pandas, NumPy, scikit-learn), SQL

AI & ML: TensorFlow, LangChain, RAG, Anthropic API, OpenAI API, Prompt Engineering

Backend & APIs: Spring Boot, Spring Framework, FastAPI, Hibernate (JPA), RESTful APIs, Microservices

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, GitLab CI, CI/CD Pipelines, Maven, Git, Linux

Databases: PostgreSQL, Oracle, InfluxDB

Testing: JUnit, Mockito, TDD, Integration Testing

Core Competencies: End-to-end Delivery, Clean Code, Agile/Scrum

Languages

English (C1 – Fluent) · German (A1 – Basic) · Malayalam (Native)